

Supplementary material

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Description. Supporting analyses: interval repair, excluded rows, a locked forecast, a stellarator-heliotron replication, specifications, robustness, identity controls, distance and bound detail, design-matrix, constraint and errors-in-variables checks, and provenance.

For: “Holding out whole tokamaks reverses tree-ensemble versus power-law rankings for energy-confinement scaling” (L. Salcedo).

This document holds the supporting analyses referred to in the main text. Sections S1 to S3 are fusion analyses whose headline numbers are stated in the main text and whose construction, tables and caveats are given here in full: repairing the interval collapse and finding the limit of the repair, replicating the ranking inversion on rows the standard analysis set excludes, and a locked forecast at three device operating points. Sections S4 to S6 are supporting material for results the main text states in prose: the three-kernel Gaussian-process ladder behind the mean-function ablation, the full model, kernel and split specification, and the per-label scores with the eligibility-threshold sweep. Sections S7 to S14 are results the main text summarises in a paragraph or a sentence each: a power law with a bounded correction; the mixed model the clustered-validation literature pairs with this split design; the choices that do not carry the reversal; the distance diagnostic in dimensionless coordinates; what substituting the boundary elongation for the areal one costs the ITPA20 laws; the rank audit of the design matrix and the refit of the published exponents; the Connor–Taylor hierarchy and a shrinkage prior in place of the constraint; and the errors-in-variables refit. Section S15 reports the external replication on the International Stellarator-Heliotron Confinement Database, specified and archived before its data were downloaded, on devices the tokamak database does not contain. Sections S16 to S18 hold material the main text states in a table or a sentence: the construction of the loss-power and device-identity controls, the distance diagnostic and the training-target bound in detail, and the provenance of every input and number.

Section and equation numbers prefixed S refer to this document; unprefixed section names refer to the main text.

Each section supports, stress-tests or documents a result or a methodological choice the main text summarises. What moved is the construction, the tables and the caveats, not the finding.

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Table S1: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out database label (LOLO) and across the ITER-size-matched cut. The device-level counterpart is Table S2.

Model	split conformal		by label		+ distance	
	LOLO	cut	LOLO	cut	LOLO	cut
random forest	35%	3%	88%	29%	88%	40%
hist grad. boosting	45%	0.3%	90%	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	95%	92%	92%
IPB98(y,2) ^{a)}	89%	88%	89%	89%	89%	87%

^{a)}Historical reference, not refitted within these folds.

Alt text. Empirical coverage of nominal 90% intervals for five models under split-conformal, by-label and distance-scaled calibration, on a held-out database label and across the ITER-size-matched cut. By-label calibration brings held-out-label coverage close to nominal for every model; coverage across the size cut remains well below nominal for the tree ensembles.

S1 Repairing the intervals, and the limit of the repair

The diagnosis in the calibrated-intervals section of the main text makes a testable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves are tested below.

The label-level scheme uses an inner leave-one-label-out loop, and nothing in it touches the target label. It is called label-level rather than machine-level deliberately: the calibration unit is the database device/wall label, so for a target of JET-ILW the calibration set still contains JET. The device-level variant that removes that overlap is reported at the end of this section. For each target m : remove m ; the remaining labels are the training set. Inside that training set, hold out each label k with at least 30 rows in turn, fit on the others, and record $|\log \hat{\tau} - \log \tau|$ on k together with each of its rows' Mahalanobis distance from that inner fit set. Pool those residuals into the calibration quantile. Then fit the final model on all the training labels and score m once. Every conformity score therefore comes from a label its own model had not seen, which is the unit the test rows differ by, and no row from m enters any fit or any quantile. The distance-scaled variant fits $\sigma(d) = \exp(c_0 + c_1 d)$ to those calibration residuals by least squares on $\log|r|$ and divides the scores by it.

Neither scheme is a finite-sample-valid group-conformal procedure, and they are not claimed as one. Two things stand in the way. The conformity score is still a row, not a label, and rows arrive in correlated clusters, so the exchangeability the split-conformal proof needs is no more available here than in the calibrated-intervals section of the main text. And in the distance-scaled arm the scale function $\sigma(d)$ is estimated from the same calibration residuals it then normalises, which the standard argument does not cover. What these are is empirical held-out-label residual calibration: the quantile is the finite-sample conformal rank applied to those scores, and the coverage reported for it is measured rather than guaranteed. Recovering a guarantee would need label-level conformity scores and a scale model fitted on separate rows or cross-fitted, and neither is done here.

Both arms behave as the diagnosis predicts. On a held-out label, empirical row-level coverage returns close to nominal: every model lands within two points of it, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. Neither recalibration procedure tested restores near-nominal coverage there, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest).

The same calibration over physical devices. The scheme above calibrates on database labels, so for a target of JET-ILW the calibration set still contains JET. Collapsing the wall variants and repeating the whole procedure over the 11 physical devices removes that overlap: training, calibration and scoring then all use the device as the unit.

Table S2: The same three schemes with wall variants collapsed, so every calibration unit is a physical device and the held-out device is genuinely unseen. Pooled row-level coverage of nominal 90% intervals, with the median multiplicative half-width in brackets.

Model	split conformal	by device	+ distance
random forest	31% (1.23 $\times$)	84% (2.46 $\times$)	88% (2.76 $\times$)
hist grad. boosting	38% (1.24 $\times$)	77% (2.29 $\times$)	80% (2.53 $\times$)
ridge, log-linear	85% (1.34 $\times$)	92% (1.40 $\times$)	93% (1.43 $\times$)
power law, collisionless	88% (1.34 $\times$)	92% (1.39 $\times$)	93% (1.42 $\times$)
IPB98(y,2) ^{a)}	90% (1.39 $\times$)	90% (1.39 $\times$)	90% (1.39 $\times$)

^{a)}Historical reference, not refitted within these folds.

Alt text. Pooled row-level coverage of nominal 90% intervals on a physically unseen held-out device, under split conformal, by-device and distance-scaled calibration, with median half-widths in brackets. Tree ensembles climb from well below nominal towards it only as interval widths roughly double, while the other models stay near nominal throughout.

Table S2 says the diagnosis holds and the repair is less complete than the label-level numbers suggest. Plain split conformal is worse on a genuinely unseen device than on an unseen label, 31% against 35% for the forest and 38% against 45% for the booster, which is what removing the within-device overlap should do. Calibrating by device then recovers most of the shortfall for the forest (84%, and 88% with distance scaling, against 88% either way at label level) but noticeably less for the booster, which reaches 77% and 80% where the label-level scheme reported 90% and 89%. The label-level number therefore overstates how well this repair calibrates the booster on a device it has never seen. Both linear models are unaffected within a point or two. The cost is visible in the widths: the trees buy that coverage at roughly 2.3 to 2.8 $\times$ multiplicative half-width against 1.23 $\times$ under plain split conformal, where the linear models stay near 1.4 $\times$.

Those coverages pool rows, and JET and ASDEX Upgrade supply most of the rows. Averaging the 11 per-device coverages instead, so that each device counts once, plain split conformal covers 23% for the forest and 43% for the booster. The collapse does not depend on the aggregation. The size of the repair does. Calibrated by device, the forest and the booster both reach 93% with devices weighted equally, against 84% and 77% pooled, so what the repair leaves uncovered lies mostly in the devices with the most rows. The three linear rows move by at most six points under any scheme.

Across the size cut with the same device grouping the picture is unchanged in kind. The forest and booster remain far from nominal at 41% and 12% by device and 57% and 21% with distance scaling, better than the label-level 29% and 4% but nowhere near 90%, while the constrained power law holds at 95% and 93%. Changing the calibration unit cannot repair a shift the calibration set does not contain, which is the point of the section.

Among the models refitted within folds, the constrained power law of the constraint section of the main text is the only one in Table S1 whose plain split-conformal coverage stays near nominal at the ITER-size-matched cut, and its intervals hold there under every scheme. IPB98(y,2) also holds at 88%, but it is a historical reference rather than a fold-fitted competitor.

S2 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis; Sec. S15 adds a second database. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against the 6228 of the deposited STD5 file analysed in the main text, and it is pinned by its own SHA-256. Matching on (tokamak, shot, time) shows that file is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 labels, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, scored against ITER89-P [1] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one failure mode that would not announce itself. If the row match against STD5 fails, every STD5 row stays in the “disjoint” arm, that arm becomes a superset of the data the ranking-inversion section of the main text was computed on, and it reproduces the ranking-inversion section of the main text by construction. The matcher is therefore checked positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% over that law reported above and computed on rows it never saw, and the best model on an unseen label is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen label is the collisionless power law of the constraint section of the main text. Counting label-model pairs where a tree ensemble loses to the published law on an unseen label: 19 of 24 and 10 of 10; against the fold-refitted power law instead, 16 of 24 and 9 of 10.

Row disjointness is not discharge disjointness, and the paper’s own argument says the discharge is the unit that matters. Of the 5358 H-mode rows, 425 (7.9%) are slices of one of 336 discharges that STD5 samples at a different time. Dropping those discharges outright leaves 4933 rows across 2864 discharges that share no shot with STD5 at all, and the arm survives: the best cross-validated model still beats the published law, by 39% against 42% on the full arm, and the best model on an unseen label is still IPB98(y,2), with the tree ensembles at 0.526 and 0.552 against the power law’s 0.332. The reversal does not depend on the shared discharges.

So the reversal is not an artefact of the standard set’s selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

S3 Locked predictions at three device operating points without matching H-mode observations

Everything above is retrospective. To make the disagreement checkable rather than merely described, what each model predicts is recorded for three real devices, with intervals, a date, and a digest over the input rows so that a later edit is visible. The parameter sets are published design values, and for SPARC and ITER, evaluating IPB98(y,2) on the adopted design vector reproduces the reference confinement time its own source quotes, to 0.6% and 2.9% respectively; the JT-60SA source supplies operating parameters rather than a quoted confinement time. Those are checks that the right machine is being described. None of the three has a *measured* H-mode thermal confinement time at the operating point tabulated here, which is what makes these predictions rather than retrodictions.

The ITER row is taken from the 1999 basis [2], and a forecast whose whole purpose is to be checkable later should say why it is locked against a document of that age. The eight engineering values it supplies for the $Q = 10$ inductive scenario, $R = 6.2$ m, $a = 2.0$ m, $I_p = 15$ MA, $B_t = 5.3$ T, $\bar{n}_e = 1.0 \times 10^{20} \text{ m}^{-3}$, $P = 87$ MW, $\kappa = 1.7$ and $M = 2.5$, are the ones the 2007 revision of the confinement and transport basis carries forward for the same scenario [3], so the choice is not a 1999 number that a 2007 number would replace. ITER’s programme has been rebaselined since, on a schedule and a staging this paper does not attempt to summarise; the 2024 baseline and the research plan built on it are described by Loarte *et al.* [4], and the row below should be read as conditional on the operating point tabulated above rather than on ITER’s current plan of record. It is not a prediction of whatever ITER’s first $Q = 10$ discharge turns out to be, and a reader who prefers a different operating point can rescore every model at one: the complete input vector is in `results/forecast.json`.

How these intervals are built. They are not the plain split-conformal intervals of the calibrated-intervals section of the main text, and deliberately so: the main text’s coverage table measures those covering 3% of rows across the ITER-size-matched cut, so quoting them for a next-step device would publish an interval already known to fail in exactly this situation. Each row instead uses the distance-scaled label-level scheme of Sec. S1, which gives the best size-cut calibration for the models that fail most severely there, applied with the whole database as the training set. Two disclosures belong with that. The scheme was chosen after comparing the three in Table S1, so these intervals are exploratory rather than specified in advance, in the same sense as the collisionless rung of the constraint section of the main text. And it calibrates over database labels rather than physical devices, so these are not device-level conformal intervals: Table S2 measures what that distinction costs, and for the tree ensembles it is not negligible. The label-level scheme is retained here because it is the one the forecast was locked under on 31 August 2026, and re-deriving the intervals afterwards would make the lock meaningless. A reader who prefers the device-level calibration should read the tree

Table S3: Predicted energy confinement time, in seconds, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest, so that the operating points these were computed at can be recovered later. Operating points are published design values: SPARC from [5] (V2 primary reference discharge), ITER from [2] ($Q = 10$ inductive baseline), and JT-60SA from its Research Plan [6] (full-current inductive scenario). Intervals are distance-scaled held-out-label calibration intervals, not the plain split-conformal intervals of the main text’s coverage table, and are pointwise intervals for a new observation; their construction is specified in the text below. The complete input vector for each device is recorded in `results/forecast.json` under the content digest quoted below.

	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2) ^{a)}	0.765 [0.58, 1.00]	0.479 [0.35, 0.66]	3.591 [2.66, 4.85]
power law, collisionless	0.724 [0.55, 0.96]	0.428 [0.31, 0.59]	2.837 [2.09, 3.84]
ridge, log-linear	0.720 [0.53, 0.98]	0.444 [0.32, 0.62]	2.858 [2.07, 3.95]
random forest	0.136 [0.03, 0.53]	0.449 [0.23, 0.87]	0.435 [0.19, 0.98]
hist grad. boosting	0.305 [0.14, 0.69]	0.418 [0.24, 0.74]	0.444 [0.24, 0.84]

^{a)}Historical reference, evaluated analytically at the operating point and not refitted; its interval is built by the same distance-scaled held-out-label calibration as the fitted rows.

Alt text. Predicted energy confinement times in seconds, with nominal 90% intervals, for SPARC, JT-60SA and ITER from IPB98(y,2) and four fitted models. Predictions agree closely for JT-60SA and diverge for SPARC and ITER, where the tree ensembles predict much shorter times than the scaling laws.

rows of Table S3 as correspondingly optimistic in width. For a given model, each eligible label k is held out in turn, the model is refitted on the remaining labels, and the absolute log residuals on k are recorded together with each of its rows’ Mahalanobis distance d from that inner fit set. A scale $\sigma(d) = \exp(c_0 + c_1 d)$ is fitted to those pairs by least squares on $\log|r|$, dropping exactly-zero residuals; the conformity scores are the residuals divided by $\sigma(d)$; and the half-width is the $[(n + 1)(1 - \alpha)]$ -th smallest such score at $\alpha = 0.10$, the same finite-sample rank used in the calibrated-intervals section of the main text. The interval reported for a device is that quantile multiplied by $\sigma(d)$ evaluated at the device’s own distance, applied symmetrically in $\log \tau$ and exponentiated. The model whose point prediction is tabulated is then fitted once on all 6228 rows. For IPB98(y,2) the same construction is used on its analytic residuals, so its interval is comparable to the others rather than merely coexisting with them. Every interval is therefore an empirical pointwise calibration interval for a new observation rather than a finite-sample-valid conformal interval. Three things stand in the way, and only the last is about ITER: the conformity scores remain clustered row-level scores, the distance scale is estimated from the calibration residuals it normalises, and every calibration unit is smaller than ITER along the size direction, which is the limitation Sec. S1 measures directly.

Table S3 illustrates the practical consequence of the disagreement in one place. On JT-60SA, which sits inside the database’s size range and is already operating, all five models agree to within 15%. On ITER, $1.82\times$ beyond it, they disagree by a factor of **8.3**.

For the random forest that gap cannot be closed by tuning the standard estimator, and the ranking-inversion section of the main text says why: it predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s, and no hyperparameter moves that ceiling. The gradient booster carries no corresponding arithmetic bound but

produces a similarly low point prediction. Because its prediction cannot exceed the largest training target, 1.321 s, the random forest cannot reach the 2.84 s to 3.59 s range the linear and physics-informed models predict for this ITER operating point, and its nominal 90% interval there runs from 0.19 s to 0.98 s, containing none of those predictions.

The JT-60SA column becomes checkable first, though not against any measurement published as of the 31 August 2026 forecast lock: the first campaign ran L-mode divertor plasmas, and its published power balance reports the energy confinement time following L-mode scaling [7], whereas the row above is an H-mode prediction scored against IPB98(y,2). Checking it needs H-mode operation at a comparable design point, which the research plan [6] places in a later campaign and which the overview of first operation and forward plans [8] discusses in view of ITER and DEMO; no date is put on that here, because the schedule is not ours to assert and the prediction is locked against a parameter vector rather than against a calendar.

Each row of Table S3 carries its own interval and is assessed separately against it, so the range quoted here names one model rather than the set. A stationary JT-60SA H-mode near this design point reporting a thermal confinement time outside 0.349 s to 0.657 s would fall outside the IPB98(y,2) reference row’s stated 90% predictive interval. The fitted rows are assessed at their own bounds, which are narrower for the constrained power law (0.313 s to 0.587 s) and wider for the random forest (0.232 s to 0.867 s). A single observation outside a nominal 90% interval is evidence against that forecast rather than a refutation of it: a perfectly calibrated model still places about one observation in ten outside such an interval, so a single miss is inconsistent with the prediction at the stated interval level and no more. Because all five intervals overlap across roughly this range, a measurement inside it separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label in the database tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

S4 The three-kernel Gaussian-process ladder

The main text states the ordering across three Gaussian-process kernels and builds its mean-function ablation on it. The ladder itself is here: the same log features, the same optimiser and the same rows, differing only in which kernel terms exist. Table S4 gives the scores, and Figure S1 shows them beside the spread of each rung’s predictions, which is where a model that has reverted to a constant becomes visible as such.

Specification. The three kernels are a squared-exponential with a single isotropic length scale shared across the standardised log features, a dot-product kernel, and their sum, each with an additive `WhiteKernel` for observation noise. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log \tau$: this is what makes “reverts to the prior” and “returns the average training target” the same statement, and the parallel with

Table S4: log-RMSE under the three splits, and the rank correlation of per-label error against extrapolation distance. The intended difference across the first three rows is whether the kernel admits a trend that continues at long range; adding the linear term also enlarges the function class. The three span a factor of ten at the ITER-size-matched cut. IPB98(y,2) is a historical reference fitted to the DB2.8 predecessor of this database [9] and is not refitted within these folds.

model	CV	held-out	ITER cut	ρ
GP, linear + RBF	0.112	0.218	0.191	-0.01
GP, RBF only	0.142	0.541	1.948	+0.65
GP, linear only	0.181	0.214	0.278	-0.06
random forest	0.128	0.465	0.938	+0.85
IPB98(y,2), historical ref.	0.199	0.188	0.194	-0.49
mean baseline	0.869	0.994	1.459	+0.56

Alt text. Table of log-RMSE under cross-validation, a held-out label and the ITER-size-matched cut, with the rank correlation of per-label error against extrapolation distance, for three Gaussian-process kernels, a random forest, the IPB98(y,2) scaling and a mean baseline. Models without a long-range trend term show error rising with distance.

a tree ensemble exact. Hyperparameters are fitted by maximising the log marginal likelihood on a seeded subsample of 1000 training rows; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold. No held-out row enters either stage. Every rung starts from the same initialisation for every term it has, so the three fits differ only in which terms exist. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term, exponentiated to τ . The kernel equations, initialisations and bounds are in Sec. S5.

The control behaves as intended: the linear-kernel process and ridge agree to 0.0000 under cross-validation and on a held-out label, and to 0.0007 at the ITER-size-matched cut, which is what licenses reading the other two rows as consequences of the kernel. The bounded rung then fails exactly as a tree does. It beats the power law under cross-validation and scores 1.948 at the ITER-size-matched cut, worse than predicting a constant, with error tracking distance at $\rho = +0.65$. The mechanism is the one in the main text: a random forest cannot exceed the largest training target, and an RBF process reverts to its prior mean. Both are structural long-range behaviours rather than tuning shortfalls, and on the held-out rows this one’s predictions carry half the spread of the truth.

Adding a dot-product term to that same kernel keeps the nonlinear RBF component it already had and supplies an unbounded linear trend alongside it. Flexibility is retained rather than held exactly fixed, since a linear kernel enlarges the function class as well as changing the asymptote; what the comparison isolates is the availability of a continuing trend. It gives the best cross-validated score in this work, 0.112 against the random forest’s 0.128; it scores 0.191 at the ITER-size-matched cut, better than the analytic law, which is a historical reference rather than a fold-fitted competitor, and second only to the constrained fit of the main text at 0.183; and its error is flat against distance at $\rho = -0.01$.

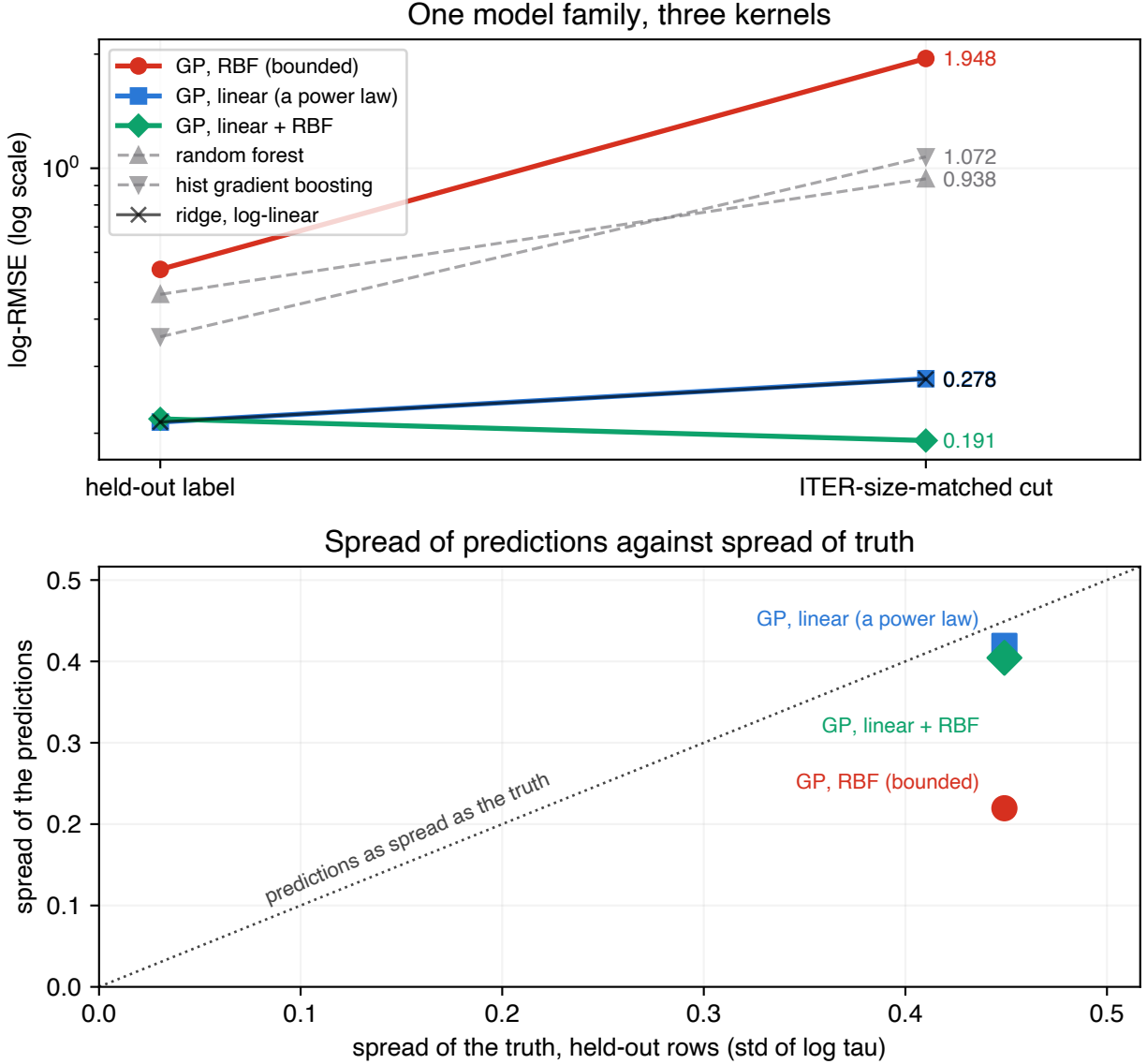


Figure S1: One model family, three kernels. Gaussian processes with a radial-basis-function (RBF) kernel, a linear (dot-product) kernel, and their sum. Top: log-RMSE on a held-out label and at the ITER-size-matched cut. The kernels differ in whether they permit a trend that continues at long range; adding the linear component also enlarges the function class. Bottom: spread of the predictions against spread of the truth on the held-out rows. A model whose posterior has reverted to its prior mean predicts nearly the same value everywhere, so its spread sits near zero however good its in-range fit was: the same failure a tree ensemble shows when it cannot exceed its largest training target, reached by a different route.

Alt text. Top: held-out error under a held-out label and the size cut for three Gaussian-process kernels and three reference models. The bounded RBF kernel degrades past the tree ensembles at the cut; the linear-plus-RBF kernel improves. Bottom: the RBF kernel's predictions spread far less than the truth, the other two about as much.

Three versions of a linear trend with a bounded correction. Three numbers in this paper answer to the description “a linear trend with a bounded correction”, and they differ, so it is worth saying how. The two trending rows of the mean-function table of the main text fix the trend first, by penalised least squares in one case and by adopting published exponents in the other, and give the residual process no say in it; they score 0.278 and 0.186 at the cut. The linear-plus-RBF kernel of Sec. 6.2 of the main text instead fits both components jointly by maximising the marginal likelihood, and scores 0.191. None of the three is a disagreement about the mechanism, which all of them exhibit. What separates them is where the trend comes from, and on this database the trend that transfers best came from outside the fit entirely. The kernel has marginally the lowest cross-validated score here, at 0.11179 against the IPB98-mean arm’s 0.11196, which is a tie at any precision worth quoting; out of distribution the two part company, 0.218 against 0.189 on a held-out label. None of the three beats the constrained fit of Sec. 6.1 of the main text at 0.183, and the two that come near it, at 0.186 and 0.191, sit inside the 0.01 resolution the Limitations section of the main text states, so they are tied with it rather than behind it. Sec. 4 of the main text reports two published laws below all of them.

What the Gaussian-process intervals are. The Gaussian-process intervals of Sec. 6.2 of the main text are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term and are exponentiated to τ ; that is the closest available analogue of a conformal interval, which is also a statement about a new observation rather than a latent mean, but the two are not the same construction and the comparison should be read as such. Those kernels are tuned by marginal likelihood on a seeded subsample of each training fold, with only the hyperparameters coming from the subsample and the posterior then conditioned on the whole fold, which is stable across the range that was affordable (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and “linear plus RBF” is one reading of “a power law with a bounded correction” among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 6.1 of the main text all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

S5 Full model, kernel and split specification

This section carries the settings the main text points at. “Library defaults” is not a specification, and a default may differ in a later release, so the consequential settings are written out rather than referenced. Library versions are pinned in the repository’s `constraints.txt` and named in the main text.

S5.1 The models of the headline comparison

The power law is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, chosen before any held-out score was computed and never varied. The random forest is `RandomForestRegressor` with 300 trees, `criterion="squared_error"`, unlimited depth, `min_samples_split = 2`, `min_samples_leaf = 1`, `max_features = 1.0` and bootstrap resampling. That last setting means every split considers all nine features rather than a random

subset, so this is bagged CART with bootstrap resampling and not the feature-subsampled forest of [10]; it is scikit-learn’s regression default, it is a choice rather than an oversight, and the nested search of the main text moves off it in 13 of 19 folds. The gradient booster is `HistGradientBoostingRegressor` with squared-error loss, learning rate 0.1, at most 100 boosting iterations, 31 leaves, no depth limit, `min_samples_leaf` = 20, no L_2 penalty and 255 bins. Its `early_stopping`="auto" resolves to *off* at every sample size here, since that setting enables early stopping only above 10 000 rows and no training fold reaches that. The polynomial controls are `PolynomialFeatures` of degree 2 and 3 without a bias column, then the same scaler and ridge. Neither tree ensemble is standardised, which is immaterial to both.

S5.2 The Gaussian-process kernels

The three kernels are

$$\begin{aligned} k_{\text{RBF}}(x, x') &= \sigma_r^2 \exp\left(-\frac{1}{2}\|x - x'\|^2/\ell^2\right) + \sigma_n^2 \delta_{xx'}, \\ k_{\text{lin}}(x, x') &= \sigma_l^2 (x^\top x' + \sigma_0^2) + \sigma_n^2 \delta_{xx'}, \\ k_{\text{lin+RBF}}(x, x') &= k_{\text{lin}} + k_{\text{RBF}} - \sigma_n^2 \delta_{xx'}, \end{aligned} \tag{1}$$

with a single isotropic length scale ℓ shared across the standardised log features, so the RBF component is isotropic rather than automatic-relevance-determining. Features are standardised inside the pipeline, so each fold’s scaler is fitted on that fold’s training rows alone and ℓ is in those units. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log \tau$: this is what makes “reverts to the prior” and “returns the average training target” the same statement, and the parallel with a tree ensemble exact. Observation noise enters as an additive `WhiteKernel` rather than through a fixed jitter. Hyperparameters are fitted by maximising the log marginal likelihood with scikit-learn’s default L-BFGS-B optimiser, `n_restarts_optimizer` = 0, from the shared initialisation $\sigma_l^2 = 1$, $\sigma_0 = 1$, $\sigma_r^2 = 0.5$, $\ell = 1$, $\sigma_n^2 = 0.05$, under the bounds $\sigma_l^2, \sigma_0 \in [10^{-3}, 10^3]$, $\sigma_r^2 \in [10^{-3}, 10^3]$, $\ell \in [10^{-2}, 10^3]$ and $\sigma_n^2 \in [10^{-6}, 1]$. Every rung starts from the same values for every term it has, so the three fits differ only in which terms exist. The optimisation runs on a subsample of exactly 1000 training rows drawn without replacement by `numpy.random.default_rng(42)`, or on the whole fold when it holds fewer; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold, which is one Cholesky factorisation. No held-out row enters either stage. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term, exponentiated to τ .

S5.3 The split-conformal construction

The construction is specified as follows, so that a reader can reproduce the coverage numbers rather than infer the procedure from them. Within each training fold, the distinct discharge identifiers are permuted with `numpy.random.default_rng` seeded at 20240811, and the first $\text{round}(0.25 n_{\text{discharges}})$ of the permutation are the calibration discharges; every row of a calibration discharge is a calibration row, and the model is fitted on the remaining rows only. Arms

with fewer than 100 calibration rows are not reported. The conformity score is the absolute log residual $|\log \hat{\tau} - \log \tau|$. The half-width is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score in ascending order, at $\alpha = 0.10$, which is the finite-sample conformal rank rather than the plain empirical quantile; the $+1$ accounts for the test point itself being one of the $n+1$ exchangeable scores, and where that rank exceeds n the half-width is reported as infinite rather than as a finite but invalid number. Ties need no rule, since the quantile is a rank on a sorted array. The interval is symmetric in $\log \tau$ and is a pointwise interval for a new observation, not for a latent mean. Every coverage in the main text’s table is pooled over rows. Grouped cross-validation pools the rows of all five folds. Leave-one-label-out calibrates and scores each held-out label on its own fold, then pools the rows of the 13 labels. The size cut pools the 2730 rows above it. The main text also gives these coverages with each held-out label weighted equally, computed from the per-label file `results/conformal_per_machine.csv`.

Holding out whole discharges keeps a calibration row and a test row from coming out of the same shot, but the conformity unit is still the row, and rows arrive in correlated clusters of quasi-stationary slices from one discharge. The standard finite-sample guarantee assumes exchangeable conformity scores and clustered rows do not supply that, so every coverage number in the main text is empirical row-level coverage rather than a guaranteed rate. A discharge-level conformity score, or a group-level conformal construction (the jackknife+ and CV+ family [11] adapted to clusters; the standard forms are not cluster-aware as they stand), would be the way to recover a guarantee, and neither is used here.

S5.4 Seeds and determinism

Every estimator carries the seed 42, and the forest is fitted in parallel but predicts single-threaded so that reruns are bit-identical. Standardisation sits inside the pipeline, so it is refitted on the training rows of every fold and never sees a held-out row. No within-model hyperparameter search is performed on any reported cross-validation score, so nested tuning is not required for those numbers, and Sec. 3.2 of the main text reports what a nested search does when one is run anyway. Comparing several prespecified model families is still a selection of a kind, and two later choices go further: the constraint rung of Sec. 6.1 of the main text and the damping of the bounded correction (Sec. S7) were both picked after seeing held-out scores, and both are flagged where they appear. The Gaussian processes of Sec. 6.2 of the main text do estimate kernel hyperparameters, by marginal likelihood on training rows alone.

S6 Per-label scores and the eligibility-threshold sweep

Both results are stated in prose in the main text. The tables are here for a reader who wants the per-unit numbers behind them: Table S5 gives the leave-one-label-out scores the distance correlation is computed over, Table S6 the sweep of the row threshold that decides which labels are scored at all, and Figure S2 the full size-ordered sweep whose matched cut the main text reports.

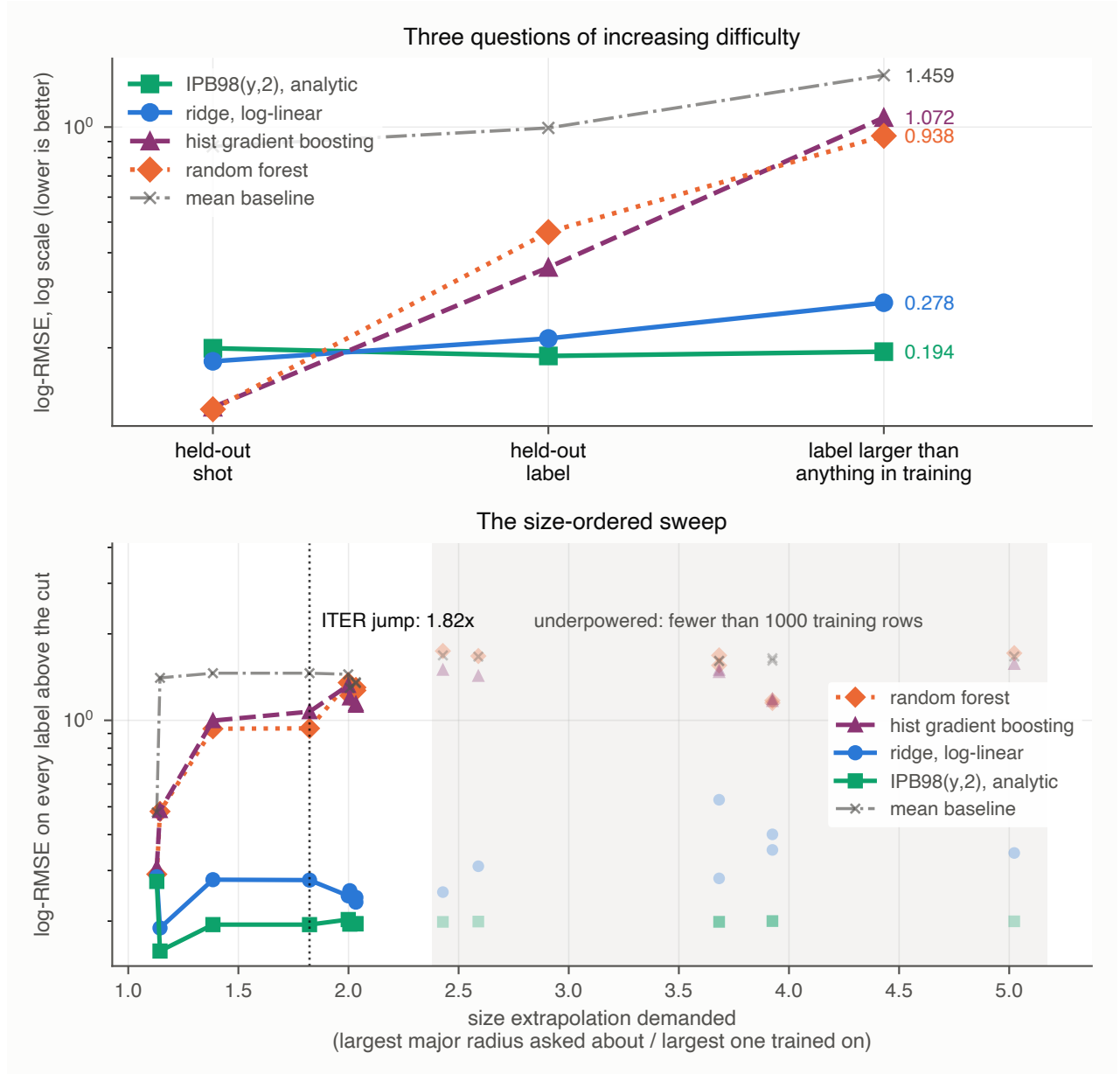


Figure S2: Size-ordered extrapolation. Top: the escalation across three splits of increasing difficulty, where the tree ensembles collapse toward the mean baseline. Bottom: the sweep across every cut, with underpowered cuts drawn faint and excluded from claims. The power law holds across the sweep and the trees do not recover.

Alt text. Top: error under held-out shot, held-out label and a label larger than anything in training. The two tree ensembles rise steeply, the ridge power law moderately, and the historical law barely at all. Bottom: across the size-ordered sweep the trees climb to the constant baseline once the demanded ratio exceeds the smallest cuts, while the power laws stay far below it throughout.

Table S5: Per-label log-RMSE under leave-one-label-out, ordered by Mahalanobis distance of the held-out label’s mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular: the design matrix is rank deficient by two, as Sec. S12 shows). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

Alt text. Table listing, for each of thirteen held-out tokamak labels, its row count, Mahalanobis distance from the training mean, and leave-one-label-out log-RMSE for the IPB98 power law, ridge, histogram gradient boosting and random forest. Tree-model errors grow with distance while power-law errors stay roughly level.

Table S6: Leave-one-label-out at four eligibility thresholds. Only the set of labels scored changes across rows; the training folds and the models are identical throughout. The 30-row threshold is the one reported elsewhere in this paper. p is the exact two-sided sign test on the win count.

min. rows	labels scored	power law	forest	forest worse on	p
10	15	0.262	0.473	14 of 15	9.8×10^{-4}
20	13	0.214	0.465	13 of 13	2.4×10^{-4}
30	13	0.214	0.465	13 of 13	2.4×10^{-4}
50	11	0.223	0.449	11 of 11	9.8×10^{-4}

Alt text. Leave-one-label-out scores at four minimum-row thresholds: labels scored, log-RMSE of the power law and the forest, the number of labels on which the forest’s error is higher, and the sign-test p -value. At every threshold the forest’s error is roughly twice the power law’s and higher on nearly every label.

S7 A power law with a bounded correction

The diagnosis in Sec. 3.1 of the main text suggests a specific repair. Fit the power law, learn a correction on its log residuals, and damp it by λ :

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}}. \quad (2)$$

At $\lambda = 0$ this is exactly the ridge row of Table 1 of the main text. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [12], and this section makes no claim on that idea. What is varied is the one property the diagnosis

points at, the long-range behaviour of the correction: one corrector is a degree-2 log polynomial, which is unbounded, and the other is depth-2 boosted trees, which never left the residual range they were trained on here. The two families differ in more than that, and the controlled version of the comparison is Sec. 6.2 of the main text. The damping factor is swept over nine values from 0 to 1, both correctors carry deliberately strong penalties so that λ rather than their own regularisation does the damping, neither corrector’s hyperparameters are tuned, and the whole estimator is refitted from scratch inside every training fold; the exact specification is in `analysis_hybrid.py` in the archived code.

Across the ITER-size-matched cut the two correctors go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out labels and by the widest margin on the most distant of them, JT-60U (0.440 to 0.281).

The mechanism is measurable. Fitted on the 14 small labels, the base power law is *biased* on the held-out ones, at a mean log residual of -0.218 , so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves its training range on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same bounded range that makes these ensembles useless as predictors of τ on a larger machine limits how much extrapolation error the correction can contribute, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero. Bounded is not the same as right: a correction confined to that range can still point the wrong way, and what is measured here is that on these rows it does not.

Two limits qualify every later summary of this section. Cross-validation selects $\lambda = 1$ in both families while the rung best on a held-out label is $\lambda = 0$, so a practitioner tuning on the split they would ordinarily use does not arrive at the rung that transfers to an unseen label. That is a statement about the held-out-label arm: across the ITER-size-matched cut the selected rung is also the best one for the bounded family, at 0.206, and the worst one for the polynomial family, at 0.356, so whether tuning on cross-validation helps or hurts there depends on the family. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, the bounded hybrid equals the power law at $\lambda = 0$, where it is that model, and beats it at all eight non-zero damping settings swept there, with the mechanism measurable.

S8 The estimator this split design is usually paired with

Every comparison above sets a pooled global fit against ensembles. The clustered-validation literature cited here for the split design [13] pairs that design with a third kind of estimator: a mixed model carrying a random intercept per device, from which a device never seen is predicted by the population-level fixed effects alone. It is neither a power law nor a tree, it is the fit matched to the deployment question, and it is scored here rather than conceded.

Write $\log \tau_{ij} = x'_{ij}\beta + u_i + \varepsilon_{ij}$ with $u_i \sim N(0, \sigma_u^2)$ over devices, fit the variance components by restricted maximum likelihood, and predict a held-out device as $x'\beta$. One quantity separates this from the pooled fit, the ratio $\lambda = \sigma_u^2/\sigma^2$. At $\lambda = 0$ the estimator *is* the pooled power law,

and it reproduces the ridge column of Table 2 of the main text to three decimals by label, at 0.214, and to two by device, at 0.212 against 0.211: the closed-form least squares this estimator reduces to and the ridge spelling of the same fit part company in the fourth decimal, which is the size of the penalty and not of a disagreement. As λ grows the intercept absorbs more of the between-device variation, which leaves β identified increasingly by variation within a device; major radius and aspect ratio are constant within a device by construction, and elongation and isotopic mass nearly so, which is exactly the identification a projection to a new machine cannot use.

The measurement follows that algebra. REML puts λ between 1.4 and 8.1 across the 13 label folds and between 2.0 and 143 across the 11 device folds, which is the spread eleven groups buy, and the held-out score tracks it: 0.264 per label against the power law’s 0.214, and **0.829** per device against 0.212, worse on 10 of the 11 devices. Because that optimum is unstable, λ is swept as well as fitted, which bounds what any rule for choosing it could deliver. Over devices the score rises monotonically with λ and is best at $\lambda = 0$, where the estimator is the pooled fit; over labels the best rung is $\lambda = 0.01$ at 0.2127 against 0.2141, a gain of 0.0014. The deployment-matched estimator therefore does not beat a pooled power law on an unseen device even when its one free quantity is chosen against the score it is being judged on. Across the ITER-size-matched cut it does help, at 0.255 against the pooled fit’s 0.279, which is the least-squares spelling of the ridge row’s 0.278, and it stays well above the constrained fit of Sec. 6.1 of the main text at 0.183 and the analytic law’s 0.194. What has been ruled out is narrower than the class: this is a single random intercept and no random slopes, so a device-level *level* shift is what fails to be the missing structure. Eleven devices is also few enough that the variance component is poorly determined, which is why the sweep rather than the fit carries the conclusion.

Random slopes are the obvious next specification and are worth saying something sharper about than “untested”. On the features that carry the size dependence they are not identifiable at all. A device-specific slope is estimated from how the response moves with that feature *within* the device, and Sec. 3.2 of the main text measures how little those features move: major radius carries 0.1% of its log variance within devices, minor radius 1.1%, inverse aspect ratio 4.9% and elongation 6.1%. There is almost no within-device variation for such a slope to be fitted to, which is the same fact that makes those columns close to device labels, read from the other side.

On the four features that do move within a device, plasma current, toroidal field, density and loss power, a random slope is identifiable, and two things stand against fitting one here. Eleven groups that already leave a single variance component spanning 2.0 to 143 will not determine a five-by-five covariance with fifteen free parameters. And the arithmetic that makes the intercept model no better than pooling on an unseen device applies unchanged: a device with no rows has no realised random effect, so it is predicted from the population-level slope whatever the random part is allowed to be, while the random part shifts the population estimate further toward within-device variation, which is the information a new machine cannot use. A richer covariance would have to help through that channel, and the sweep in this section already bounds what that channel can deliver. The specification and the full sweep are in `analysis_mixed_model.py` and `results/mixed_model.json`.

The win counts support an exact test, under a null that is worth naming precisely. Call it the independent-unit-sign null: each unit, label or device as the case may be, is assigned

to either model independently with equal probability. Under it, losing 13 of 13 has two-sided $p = 2.4 \times 10^{-4}$ and losing 11 of 11 has $p = 9.8 \times 10^{-4}$. The device-collapsed count is the one to quote, since the 13 database labels are not 13 independent devices, and the same non-independence is a reason to prefer these counts to the 13-label bootstrap.

That null is an idealisation, and Sec. 2 of the main text says in what way: any two of these fits share all but two of their training machines, so the outcomes are correlated through the training set even where the held-out devices are distinct. The descriptive statement carries the claim on its own. The forest is worse on every physical device scored, and the paired interval on the per-device difference excludes zero.

S9 Choices that do not carry the reversal

The labels too small to score at all. Five labels hold fewer than 30 rows (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them, under 1% of the database) and none can carry a held-out score of its own. They are nonetheless in every training fold, since the threshold governs scoring rather than training, and they are among the most unusual points in the feature space, so whether a flexible model can exploit those few rows is a fair question to ask of the ranking.

Dropping them from training as well answers it. Every fitted model gets slightly worse without them, by 0.012 for the power law, 0.027 for the gradient booster and 0.007 for the random forest, and the ordering over the 13 held-out labels is unchanged: power law 0.214 to 0.226, gradient booster 0.359 to 0.387, random forest 0.465 to 0.472. The 43 rows help the tree ensembles slightly, as one would expect of rows in a sparse region, and nowhere near enough to recover the inversion. The analytic reference is unaffected, since it fits nothing.

Where the row threshold is set. The 13-of-13 count is reported at a 30-row eligibility threshold, which is a choice, so it is swept. Sec. S6 scores leave-one-label-out at 10, 20, 30 and 50 rows. Only which labels are *scored* changes: every label stays in every training fold at every threshold, so the models being compared do not change. The power law wins every scored label at 20, 30 and 50 rows, and 14 of 15 at 10 rows, where the exception is TCV. TCV carries 11 rows, which is why it sits below the threshold in the first place, and a per-label log-RMSE over 11 rows is not a quantity this paper would report on its own. The mean gap moves between +0.211 and +0.251 across the four thresholds and the sign test stays below 10^{-3} throughout, so the headline does not depend on where the line is drawn.

One partition, or any partition. The cross-validation column comes from a single deterministic `GroupKFold` assignment of discharges to five folds. Since this paper is about validation protocol, whether the 29% margin is peculiar to that one partition is a fair question. Repeating the comparison over ten shuffled assignments of the same discharges into the same number of folds gives a mean forest advantage of 27.8% over the power law, ranging from 27.0% to 28.5%, with the forest ahead in all ten. The deterministic partition's 29.5% sits just above that range, so the reported margin is if anything mildly favourable to the forest, and the interpolation win it later gives up is a property of the comparison rather than of one fold assignment.

One ordering inside that sweep is not resolved and the main text says so. The two ensembles

are separated by 0.002 in Table 1 of the main text, and across these ten partitions the forest is ahead of the gradient booster in only 2 of 10, and the deterministic partition is one of the two. What every partition agrees on is the power law’s position, last of the three under cross-validation and first under leave-one-out, which is the comparison the paper’s argument rests on. Which of the two ensembles is nominally best under cross-validation is a coin the partition tosses, and no claim here turns on it.

Tuning the flexible models. Every score above uses the prespecified, untuned settings of Sec. 2 of the main text, which leaves open whether the ensembles lost only because they were not hyperparameter-tuned. That is tested by giving both a grid search nested inside each training fold: three grouped inner folds by discharge over the training rows only, so the held-out discharge fold, label or size cut never takes part in choosing a hyperparameter. The forest searched `max_features` over $\{1.0, 0.5, \text{sqrt}\}$ and `min_samples_leaf` over $\{1, 5\}$, the booster `learning_rate` over $\{0.05, 0.1\}$ and `max_leaf_nodes` over $\{15, 31\}$.

The search does real work on the forest and changes nothing about the conclusion. It moved away from the baseline `max_features = 1.0`, `min_samples_leaf = 1` setting in 13 of the 19 folds, and it helps where the paper says a flexible model should help: cross-validation improves from 0.128 to 0.126 and the leave-one-label-out mean from 0.465 to 0.403. At the ITER-size-matched cut it makes the forest *worse*, 0.938 to 1.121, against the power law’s 0.278. The booster’s search returned scikit-learn’s defaults in all 19 folds, so its numbers are unchanged.

The inner folds above group by discharge, which is how a practitioner would ordinarily tune. Grouping them by whole machine instead matches the way the model is deployed, and it helps both models: the forest falls to 0.376 and the booster to 0.350. Both remain well above the power law’s 0.214, so the reversal survives either selection procedure.

Tuning therefore widens the interpolation margin that later inverts and does not repair the extrapolation failure, which is what the training-target bound predicts: no setting of these hyperparameters moves a ceiling that comes from averaging training targets.

The penalty on the power law. The headline power law is a ridge regression with scikit-learn’s default penalty, $\alpha = 1$ on standardised columns, where confinement scalings are historically fitted by unpenalised least squares. `analysis_ols_baseline.py` scores the unpenalised fit, the unconstrained rung of the Connor–Taylor hierarchy, beside the two tree ensembles under both leave-one-out splits. The two power laws are indistinguishable: 0.214 against 0.214 by label and 0.212 against 0.211 by device, a mean difference of 0.0001. Against the unpenalised fit the forest loses on 13 of 13 labels and 11 of 11 devices, at mean gaps of +0.251 and +0.315, and the booster on 12 of 13 and 10 of 11: the same counts as against ridge. The penalty is not what the reversal rests on.

The reconstructed minor radius. The nine engineering features carry one exact dependency, $a = \epsilon R$, which a tree can split on separately. `analysis_eight_features.py` scores the three primary models on the eight independent features, with the minor radius dropped, under every split of the main text. Nothing moves that matters: the forest’s cross-validation score is 0.128 on both sets, it loses to the power law on 13 of 13 labels and 11 of 11 devices at mean gaps of +0.240 and +0.304 against +0.251 and +0.315 with nine, and across the

ITER-size-matched cut it scores 0.973 against 0.938 with nine, where the power law scores 0.278 on either set. The redundant column neither helps nor hurts the ensembles.

Two conventions this paper does not follow. Neither the population nor the weighting used above matches how the ITPA scalings were produced, and both are choices rather than findings, so both are measured.

IPB98(y,2) was fitted on a selection that excluded spherical tokamaks; this database does not, and MAST and NSTX are two of the 13 scored labels and two of the four most distant. The main text states the outcome of this control and of the same control at the ITER-size-matched cut; the construction is here. Dropping every label whose median inverse aspect ratio exceeds 0.5, which removes MAST, NSTX and START and 232 rows, from training and scoring alike, leaves the reversal intact: the forest is worse than the power law on all 11 remaining labels, mean gap +0.182, exact $p = 9.8 \times 10^{-4}$, at 0.395 against the power law's 0.212. The inversion is not an artefact of including the spherical devices.

The weighting arm is the one that qualifies the headline. Every fit above weights rows equally, so JET and ASDEX Upgrade, which supply 77% of them, set most of what each model learns. Refitting every model with each label weighted equally hurts the power law more than the forest, 0.214 to 0.259 against 0.465 to 0.447, and the win count falls from 13 of 13 to **12 of 13** ($p = 3.4 \times 10^{-3}$, mean gap +0.188). The label that changes hands is AUG-W, whose unweighted margin was the second narrowest of the thirteen at +0.012, behind JT-60U's +0.006 which does not change hands, and which moves to -0.067. The direction and its significance survive the reweighting; unanimity does not, and the 13 of 13 quoted throughout is the unweighted count.

S10 The same distance in dimensionless coordinates

Organising anything in this field by engineering variables invites one objection, and it is the objection Hall *et al.* [14] have raised in a conference presentation: the weak size dependence in this database may come from a subset localised in *dimensionless* space rather than from size, in which case a distance in engineering space measures a shadow of the thing that matters. That is checkable here rather than only citable. The Connor–Taylor constraints of Sec. 6.1 of the main text are derived in code from the scalings of ρ^* , β and ν^* , so the group definitions are already written down, and the stored energy joined across in Sec. S16 supplies the temperature they need. Placing every row by

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \quad \beta \sim n T B^{-2}, \quad \nu^* \sim n L T^{-2}, \quad (3)$$

with $T \sim W_{\text{th}}/(nV)$ and $V \sim R a^2 \kappa$, and repeating the distance construction above on those three coordinates instead of the nine, gives a second ranking of the 13 labels. Multiplicative constants are dropped throughout and cannot matter, since every quantity here is a difference of mean log-vectors.

The two rankings agree at $\rho = +0.78$, and the diagnosis does not depend on which is used. On the 6214 rows a dimensionless placement is possible for, the forest's per-label error tracks dimensionless distance at $\rho = +0.77$ against +0.79 for engineering distance, the booster at +0.35 against +0.51, and the power law is flat against both, -0.04 and -0.06. The headline

+0.85 is the engineering figure on all 6228 rows; dropping the 14 rows with no stored energy moves it to +0.79, and the coordinate change then moves it by 0.02. The distance reading above survives translation into the coordinates the objection is stated in.

The size cut does not fare as well, and this is where the objection lands. At the ITER-size-matched cut the two sides are separated in dimensionless space as well as in size: the interquartile ranges of $\log \rho^*$ do not overlap at all and its median moves by 1.21 training standard deviations, $\log \nu^*$ overlaps by 0.26 at 0.85 standard deviations, and only $\log \beta$ stays put, overlapping by 0.69 at 0.22. Larger machines in this database run at smaller normalised gyroradius and lower collisionality, which is physics rather than an artefact, and it means the cut is not a size boundary with everything else held fixed. Every result scored across it is therefore transfer across a joint displacement in size, ρ^* and ν^* , and the paper says so rather than describing it as a size extrapolation alone. The leave-one-device-out comparison of Sec. 3 of the main text carries no such confound, which is the second reason it and not the cut is the claim this paper rests on.

S11 What the elongation substitution costs

ITPA20 and ITPA20-IL are written on the areal elongation $\kappa_a = S/(\pi a^2)$, the variable IPB98(y,2) uses and the one STD5 delivers as KAPPAA [9], and Sec. 4 of the main text scores all three laws on that column. The full DB5.2.3 revision also carries the boundary elongation κ , the plasma's height over its width at the separatrix, and on most rows the two are close enough that one is easily taken for the other. This section measures what taking the boundary column for the areal one would cost the two newer laws. The measurement is also the argument for not modelling it.

On a parametrised boundary κ/κ_a depends on the triangularity alone, which confines the ratio to [1.000, 1.110]. The measured ratio has a median of **1.083** and a maximum of **1.341**, with 16% of rows above that ceiling and 7% *below* unity, which no boundary of that family produces at all. Only **2.0%** are below it by more than 5×10^{-4} ; the rest is rounding on machines that ran circular.

The rows below unity are worth following, because the obvious reading of them is that the delivered column is wrong and the obvious reading is not right here. A boundary elongation below the areal elongation is impossible for a *convex* boundary, and of the 2.0% that clear rounding, all but PBX-M's 59 rows sit within 3.5% of unity on near-circular machines, which the two columns' precision covers. PBX-M does not: its ratio runs from 0.67 to 0.80, at a median of **0.739**. PBX-M ran indented, bean-shaped plasmas, its boundary is not convex, and the convexity argument does not apply to it. DB5.2.3 records that directly: of the analysed rows, PBX-M is the only label carrying an indentation above 0.04, at a median of **0.154**, and the 60 rows with any indentation at all have a median κ/κ_a of 0.740. The column is not defective on those rows; the parametrised boundary is simply the wrong shape for them, which is the same point the ratio makes everywhere else and is here checkable against a third column.

Scoring on the boundary column rather than the areal one moves ITPA20 by up to 0.018 and ITPA20-IL by up to 0.022, all of it for the worse: on every arm scored here each law fits the variable it is written on better than the substitute, which is the direction that confirms which column that is. The shift is twenty to thirty times what the shape model would predict.

Where a column exists, a model of it is not evidence about it.

The three published laws on the rows analysed here. Evaluated analytically on the rows analysed here, ITPA20 scores 0.181 over all rows, 0.173 per label and **0.165** across the ITER-size-matched cut; ITPA20-IL scores 0.226, 0.247 and **0.158**, against IPB98(y,2)’s 0.199, 0.188 and 0.194. Both are below the power law refitted inside these folds at that cut, which scores 0.278, and against the random forest’s 0.938 and the gradient booster’s 1.072 there ITPA20 is better by factors of 5.7 and 6.5. Neither law is blind to the held-out rows and neither is offered as blind: both were fitted to selections out of DB5.2.3, so at the size cut they had seen the large machines the fitted models are being asked to reach. What it carries is that no published power law in this family lands anywhere near where the tree ensembles land, and that the two fitted to this revision are the two that beat a refit of the same functional form on these rows.

S12 The design matrix and the refit

Three column counts appear in the main text and they are not the same set. The models see nine log engineering features. Adding the analytic IPB98 prediction gives the ten-column matrix examined here, and the refit of Eq. (1) of the main text in Sec. S12.1 drops both redundant columns and uses nine, eight independent variables plus an intercept, which is why its condition number is a healthy 10.7.

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14} . Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16} : minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prediction is a fixed log-linear combination of the other eight features. The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit.

Neither the redundant column nor the estimator is load-bearing. Dropping `log_a_m` and refitting on eight independent columns leaves the log-linear scores unchanged to four decimals under all three splits, moves the tree ensembles by at most 0.035, and changes no ordering anywhere. Ordinary least squares on those same eight columns reproduces the reported ridge to six decimals under cross-validation and to 0.0012 at the size cut, so the ridge here is a numerically stable spelling of the least-squares power law rather than a regularised alternative to it.

S12.1 Refitted exponents and a weakly identified direction

Refitting Eq. (1) of the main text on all STD5 rows (design matrix 6228×9 , condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. The disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks, and a refit on a different population is a different regression. It is not a solver artefact either; independent implementations by the normal equations, by QR and by the SVD pseudoinverse agree to 7.6×10^{-13} . Decomposing the difference between the exponent vectors along the

singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix’s variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, and it is weak for a structural reason: tokamaks are not designed to vary the two independently, so more shots at the operating points these machines already run would not separate it, though shots deliberately placed to break the correlation could.

Why the size dependence in this database is weaker than IPB98(y,2)’s is itself an active question, and Hall *et al.* [15] attack it directly on DB5.2.3, treating the multicollinearity among the predictors explicitly and searching for the smallest subset of points that most reduces the fitted major-radius exponent. A continuation [14] argues, in a conference presentation, that the weak size dependence is attributable predominantly to a subset of discharges localised in dimensionless space, distinguished by normalised pressure, collisionality, normalised gyroradius and safety factor rather than by any single device or by metallic walls, and reports that excluding it recovers a major-radius dependence broadly consistent with IPB98(y,2). That is a different question from the one asked here, and the two bear on each other through the size-ordered cut of Sec. 4 of the main text, which measures transfer across a boundary those authors would draw differently. That is a caveat on how much weight that one axis can carry, and not on the ranking reversal, which is measured per unit rather than per size.

S13 The Connor–Taylor hierarchy, and a prior in place of the constraint

Sec. 6.1 of the main text imposes the Connor–Taylor similarity constraints as linear equalities on the exponents, one row of C per admissible scale transformation. Table S7 sets out the hierarchy it uses.

Table S7: The Connor–Taylor hierarchy. Each row fixes a set of dimensionless groups and contributes one linear equation in the exponents per admissible scale transformation, so fixing fewer groups imposes more constraints.

constraint	groups held fixed	rows of C
Kadomtsev [16]	ρ^*, β, ν^*	1
collisionless	ρ^*, β	2
electrostatic	ρ^*	3

Alt text. Table listing the three Connor–Taylor similarity constraints, Kadomtsev, collisionless and electrostatic, with the dimensionless groups each holds fixed and the number of linear equations each contributes to the exponent fit. Holding fewer groups fixed admits more scale transformations and so imposes more constraint rows.

Fixing fewer groups admits more transformations and therefore imposes more constraints, which is why the hierarchy runs the way it does. C and d are built by `dimensional.constraint_matrix` from Eq. (3) of the main text alone, and their numerical rows are in `results/dimensional.json`.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. S12.1, is worth

much less. That comparison sweeps three shrinkage geometries (isotropic, spectral and truncated) over the same nine penalty strengths $10^{-3}, 10^{-2}, \dots, 10^5$, tabulated in full in `results/dimensional_prior_sweep.csv`. Targeting the shrinkage beats isotropic shrinkage at every one of those matched penalty strengths, which is a real effect, but nothing in that family beats simply adopting IPB98’s published exponents outright. The hard constraint used here restricts the fit to a surface the Connor–Taylor similarity assumptions permit, where the Gaussian prior tested here favours a location and a direction in parameter space, and in this experiment the surface transfers better. That is a statement about these two families rather than about priors in general: a prior can favour a subspace, and approaches a hard constraint as its variance goes to zero.

Where the newer published laws sit on these surfaces. The same machinery places the published laws on the surfaces. The two that transfer best across the ITER-size-matched cut (Sec. 4 of the main text) are also the two furthest from the Kadomtsev and collisionless surfaces derived here. Their triangularity term is dimensionless and contributes no column to C , so the eight exponents the constraint is written over compare directly.

Table S8: Distance from each Connor–Taylor surface, as $\|Cb - d\|$ over the eight exponents the constraint is written over, beside the score each set of exponents gives across the ITER-size-matched cut. The two published laws fitted to this database revision are the two furthest from the Kadomtsev and collisionless surfaces and the two best across the cut. The numerical rows are in `results/dimensional.json`.

exponents	Kadomtsev	collisionless	electrostatic	ITER cut
IPB98(y,2), published	0.00096	0.0045	0.501	0.194
free refit (this database)	0.0031	0.064	0.817	0.279
ITPA20, published	0.0147	0.089	0.745	0.165
ITPA20-IL, published	0.0142	0.106	0.853	0.158

Alt text. Table giving, for four sets of scaling-law exponents, the distance from three Connor–Taylor surfaces alongside the score across the ITER-size-matched cut. The two published ITPA20 laws lie furthest from the Kadomtsev and collisionless surfaces yet score lowest at the cut, so proximity to a surface does not order the four.

Proximity to a surface therefore does not order these four, and no claim here rests on reading it that way. ITPA20 sits fifteen times further from the Kadomtsev surface than IPB98(y,2) and scores 0.165 against 0.194 at the cut, and the free refit sits closer to that surface than either ITPA20 law while scoring 0.279, the worst of the four. A small residual is not a predictor of transfer on this evidence.

What the hierarchy measures is narrower than that, and it is measured the only way a constraint can be: against a refit on these rows where the constraint is the single difference, the same nine features, the same closed-form solver and no hyperparameter separating 0.279 from 0.183. Table S8 is about four different fitting populations and cannot weaken a comparison internal to one.

It does point at what else a fitted law has available, and that is this paper’s own argument arriving from the fitting side rather than the scoring side. ITPA20-IL is fitted to an ITER-like selection out of DB5.2.3 rather than to the whole of it, which is what its name denotes. The

main text asks that a validation split match the deployment being claimed; restricting the fitting population to the regime being extrapolated *to* is the same principle applied to the fit rather than to the score, and across this cut it is worth more than any constraint tested here. The two are not competing repairs. A constraint is what a fit can carry when the population cannot be restricted, which is the situation of anyone refitting the published database as this paper does; the ITER-like selection is a choice the ITPA authors could make and this analysis could not, since the machines defining that regime are the ones held out. Both supply structure that the rows alone do not, which is the section’s claim, and neither is evidence that the other is unnecessary.

Per-device counts for the constrained fits. The Kadomtsev rung owes a count as well as a score. This paper reports a per-unit win count wherever it criticises a model, and the same statistic is available for the models it recommends: over the 11 physical devices the Kadomtsev fit is worse than the unconstrained one on **7** of them, at a mean paired difference of 0.001 in its favour, and the collisionless fit is worse on **5**, at 0.009 (`results/device_arm.json`). Neither count is evidence that a constraint hurts, and the last row of the constraint-hierarchy table of the main text is the calibration that says why: the unconstrained fit scored against the ridge spelling of the same power law, two solvers for one model separated by 0.00008 per device, is worse on 8 of the 11. At the gaps separating the rows of that table a per-device count is a coin toss, and none of them carries what the 11 of 11 of Sec. 3 of the main text carries. What survives the observation is the size sweep, where the gaps are an order of magnitude larger: the honest summary is that a constraint pays where the training half is small or shifted, and sits inside the resolution of the Limitations section of the main text where it is not.

Every constrained fit scored slightly worse than the unconstrained one under the cross-validation used here, so CV supplies no signal pointing at the constraint that transfers best. That is an observation about these folds rather than a theorem: a hard equality constraint can only match or worsen the *training* residual, and nothing forces it to worsen a held-out score.

S14 Errors in variables

This section gives the refit behind the statement in the Limitations of the main text that no errors-in-variables fit is offered. Every power-law fit here treats the engineering parameters as measured without error, which the confinement-scaling literature has objected to for as long as these fits have been published, because the predictors carry real measurement uncertainty. Kardaun [17] treats the problem directly, Cordey *et al.* [18] refit the database under it, and Verdoolaege *et al.* [19] give a geodesic least-squares regression built to be robust to exactly the noise an ordinary least-squares fit assumes away. None of the three is attempted here. Refitting Eq. (1) of the main text by orthogonal distance regression, which spreads the error across the predictors instead, moves the exponents enormously: up to 5.6 on the inverse aspect ratio, with the major radius going from 1.58 to 5.40, and it more than doubles the in-sample error, 0.454 against 0.181. Both are symptoms of one thing rather than a better fit. With no per-variable measurement uncertainties published for this database, an errors-in-variables estimator has nothing pinning it down along the weak singular direction of Sec. S12.1, which

carries 0.3% of the design’s variance, and it runs away along exactly that direction. So these are least-squares exponents, an errors-in-variables fit would differ, and what is reported here is a fact about the naive estimator rather than an answer to what a credible measurement-error fit would give.

S15 External replication on the stellarator-heliotron confinement database

Every fusion result above comes from one ITPA database family, and a reader can ask whether the ranking inversion belongs to that compilation. This section puts the same comparison to a database assembled by a different collaboration, on devices the tokamak database does not contain, under rules that were written down and archived before the file was downloaded. The question it answers is the one the main text asks: when the same broad model comparison is moved from validation within known devices to prediction of an entirely held-out device, how does model performance change? The outcome was to be reported whichever of five predeclared verdicts it landed on, one of which is that the database cannot be evaluated and another that it contradicts the original result.

S15.1 Dataset and prospective specification

The International Stellarator-Heliotron Confinement Database (ISHCDB) is the stellarator counterpart of the ITPA confinement database: the same IEA lineage, column conventions inherited from it, and the database the ISS04 scaling [20] was fitted to, as IPB98(y,2) was fitted to the ITPA one. Version 26 is public, in an open directory on the IPP Greifswald server (<https://ishpdb.ipp-hgw.mpg.de/ISS/>), so it is fetched by a script, pinned and verified on load like the other deposits, and never redistributed. The analysed file, `ISHCDB_26.txt`, is 3311757 bytes with SHA-256

3a9f857b768ebd601385a9d8e25ac7df4d48b3403795a43dc484ba6b17fc5f55

taken from two independent downloads that agreed. It was assembled separately from the ITPA database and shares no device with it. It is not unrelated to it: both grew under the same IEA programme, and the column conventions of one are the other’s.

The rules were fixed in `docs/ishcdb-replication-lock.md` at commit `c5390bc`, written from the database’s own documentation before the file was downloaded, and the repository’s history and the host’s push times record the order that followed: the pipeline built on synthetic data, the file’s digest committed before the file was opened, one amendment, the analysis plan frozen and pinned at commit `7a0f179`, and the analysis run once. The amendment (commit `d8afd9b`) is in the lock’s own log. The lock as first written named Heliotron E as the one device whose standard target is the thermal rather than the diamagnetic confinement time; the documentation says the same of TJ-II, in a section added with that device’s data, and the file carries no diamagnetic time for TJ-II at all. The change rests on the documentation and on missingness, and was made after the file’s columns had been counted and before any model was fitted. Every threshold below was fixed in the lock.

The primary population is the database’s standard set, the counterpart of the ITPA standard analysis set STD5, on measured rows only: the documentation calls the rows labelled

W7-X and ITER predictive data, so they leave before anything else happens, training included. The manuscript’s nine tokamak features do not apply, since a stellarator has no plasma current to scale with; the regressors are the six of ISS04, and whichever of them the file supports on 90% of rows, decided from missingness alone, is the frozen set. A device is scored if it has at least 30 rows after cleaning, the manuscript’s own threshold, with 10 as a sensitivity.

S15.2 Target and feature mapping

Table S9: ISS04’s regressors and the confinement time, as ISHCDB delivers them. The rotational transform is present on 39% of the standard set, so the 90% rule fixed in advance dropped it for every model, and the frozen set has 5 of the 6 regressors.

Quantity	column	conversion	status
confinement time	TAUEDIA, or TAUETH for Heliotron E and TJ-II	s, as delivered	target
effective minor radius	AEFF	m, as delivered	used
major radius	RGEO	m, as delivered	used
absorbed power	PTOT	W to MW	used
line-averaged density	NEBAR	m^{-3} to 10^{19}m^{-3}	used
toroidal field	BT	T, absolute value	used
rotational transform $\iota_{2/3}$	IOTA23	absolute value	omitted, finite and positive on 39% of rows, below 90%

Alt text. Table mapping the confinement time and the six stellarator scaling regressors to their ISHCDB columns, with each column’s conversion and status. The confinement time is the target, five regressors are used as delivered, rescaled or in absolute value, and the rotational transform is omitted because too few rows carry it.

Table S9 gives the mapping. The target follows the standard set’s own convention: the diamagnetic confinement time, and the thermal one for the two devices whose standard set is defined on it. ISS04 is $\tau = 0.134 a^{2.28} R^{0.64} P^{-0.61} \bar{n}^{0.54} B^{0.84} \iota_{2/3}^{0.41}$, with P in megawatts and $\bar{n}$ in units of 10^{19}m^{-3} . Because the frozen set lacks the transform, the published law cannot be evaluated on it, and this replication has no analytic reference outside the ranking where the main text has IPB98(y,2). Nor is there a feature-matched rerun of HDB5: the tokamak set has no counterpart of $\iota_{2/3}$, so the two would not be the same experiment, and the comparison between the databases is of the pattern.

S15.3 Population

Table S10 counts the rows. The file delivers 4940 rows. Dropping the 76 predictive rows leaves 4864 measured rows over 9 devices, of which the standard set holds 1719, all complete on the frozen features, over 8 devices, in 1450 discharges; 181 discharges contribute more than one

Table S10: Rows per measured device: as delivered, in the standard set, complete on the frozen features, and whether the device is scored at the 30-row threshold. HSX has no standard-set row, which the documentation explains: its stored energy is thought to be largely nonthermal and it was excluded from ISS04. W7-A’s 13 rows stay in every training fold.

Device	rows delivered	in the standard set	complete	scored
W7-AS	1156	629	629	yes
TJ-II	1132	316	316	yes
ATF	234	229	229	yes
CHS	196	196	196	yes
LHD	1362	162	162	yes
HELE	121	120	120	yes
HELJ	111	54	54	yes
W7-A	13	13	13	no
HSX	539	0	0	no

Alt text. Rows per stellarator device delivered, in the standard set and complete on the frozen features, with whether each is scored. Every standard-set row is complete, the standard set roughly matches the delivered count for some devices and is much smaller for others, and seven devices are scored while two are not.

time slice, the largest 7. Seven devices clear the 30-row threshold. None of the seven is in HDB5.

S15.4 Validation protocol

The protocol is the main text’s, run by the same code. Cross-validation is `GroupKFold` with 5 folds grouped on device and discharge, so every slice of one discharge falls in one fold. The held-out unit is the device as the database names it, with no merging. Both splits are summarised the same way: a log-RMSE per eligible device, from out-of-fold predictions under cross-validation and from held-out predictions under device holdout, averaged with each device counting once, so that the split is the only thing that differs between the two columns. The models, their settings and their seed are those of Sec. S5, untouched, with no tuning on this database for any reason. Distance is the Mahalanobis distance of a device’s mean log-feature vector from its training rows, as in the main text. The paired interval is the percentile bootstrap over devices, 2000 resamples seeded at 20240617, and the win count carries an exact sign test, both read as the main text reads them. The verdict rule fixed in the lock counts a reproduction only when the flexible model wins cross-validation, loses device holdout on average and on more than half the devices, and degrades at least 1.5 times as much as the power law does, measured by D , the ratio of the two models’ holdout-to-cross-validation ratios; a crossing smaller than that falls through to “no important differential degradation”.

S15.5 Primary result

Table S11 scores every model under both splits. On the standard set the random forest is better than the log-linear power law under cross-validation, 0.209 against 0.288, and worse on every one of the 7 held-out devices, 0.814 against 0.442 on average: worse on **7 of 7**, mean

Table S11: Each model under both splits. Pooled CV is one log-RMSE over every row; matched CV and device held out are per-device scores averaged with each device counting once, and the ratio is the second over the first. The rank columns of the main text are not repeated because the ordering is visible: on the standard set the three primary models trade places exactly as in Table 1 of the main text. The three ranked models are also reported on every measured row, a secondary population that adds HSX, LHD’s high-beta survey and two W7-AS rows whose delivered confinement times run to thousands of seconds, which the lock’s rule of no outlier removal keeps.

Population	Model	pooled CV	matched CV	held out	ratio
standard set	ridge, log-linear	0.255	0.288	0.442	1.53×
	random forest	0.182	0.209	0.814	3.89×
	hist gradient boosting	0.178	0.203	0.950	4.68×
	mean baseline	1.038	0.993	1.090	1.10×
	ridge, log-quadratic	0.220	0.242	0.417	1.72×
	ridge, log-cubic	0.198	0.220	1.009	4.59×
	GP, linear + RBF	0.167	0.195	0.409	2.10×
every measured row	ridge, log-linear	0.323	0.361	0.699	1.94×
	random forest	0.188	0.223	0.688	3.09×
	hist gradient boosting	0.259	0.244	1.041	4.26×

Alt text. Logarithmic root-mean-square error on the stellarator database under pooled cross-validation, matched cross-validation and device holdout, with the holdout-to-cross-validation ratio, for the standard set and for every measured row. The tree-based models score lower than the log-linear power law under cross-validation and show larger holdout-to-cross-validation ratios in both populations.

gap +0.373 with a paired interval of [+0.106, +0.797], and $D = 2.54$. The gradient booster is worse on 7 of 7, mean gap +0.508 [+0.206, +0.899], $D = 3.05$. Under the predeclared rule both are a reproduction of the inversion. The exact sign test returns $p = 0.016$ for each, which is the smallest value seven devices can return, and the intervals are wide; the counts and the sizes of the gaps carry the result, as in the main text. The linear-plus-RBF Gaussian process again scores lowest under both splits, 0.195 and 0.409, and again does not reverse, so cross-validation’s choice does not collapse here either once that model is in the candidate set. LHD is the hardest device for every model, at the largest distance from the rest: the power law scores 0.790 on it and the forest 2.378.

S15.6 Extrapolation distance

Figure S3 draws both splits and the distance diagnostic. This is the one part of the main text’s pattern that did not reproduce. On HDB5 the forest’s per-label error tracks distance at $\rho = +0.85$ and the power law’s does not, at $\rho = -0.06$. Here both track it: the forest at $\rho = +0.64$ and the power law at $\rho = +0.57$, over seven points, with the Gaussian process flattest at $\rho = +0.21$. Sec. S15.8 says why the power law’s error rises with distance here and did not there.

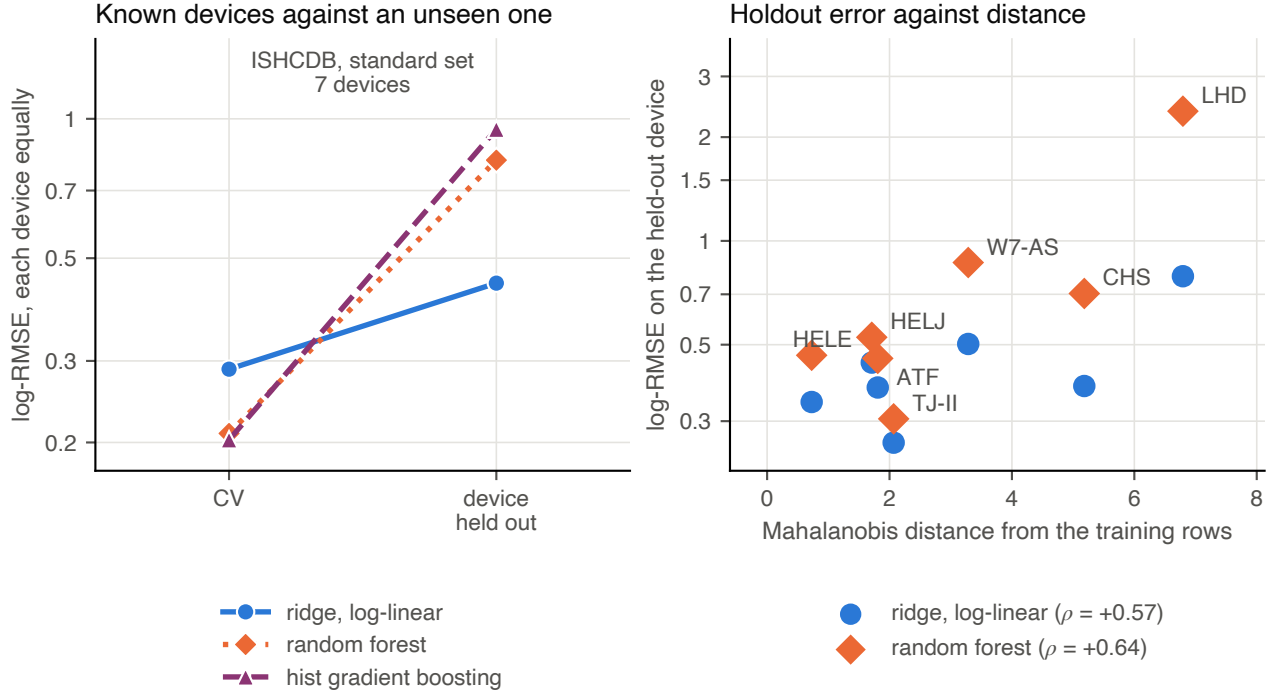


Figure S3: The replication on the stellarator database. Left: each of the three primary models under both splits, each device counting once, on the standard set. Right: per-device holdout error against the Mahalanobis distance of the device from its training rows, for the power law and the random forest, with the rank correlation over the seven devices in the legend. The correlation is described and not tested at this device count.

Alt text. Left: on the stellarator database the power law’s error rises moderately from cross-validation to a held-out device, while the forest starts lower and rises above it, so the lines cross. Right: per-device error rises with distance from the training rows for both models, with LHD the furthest and the worst.

S15.7 Devices absent from HDB5

Every scored device is absent from HDB5, so the descriptive subgroup that the tokamak replication of Sec. S2 could not supply is here the whole result. The two databases share a lineage and a schema and no machine.

S15.8 The offset the lock predicted

Table S12: Each ranked model’s error under both splits, whole and with each device’s mean log residual removed. Per device the mean squared residual splits exactly into the squared mean and the variance about it; removing the mean grants the model a free constant on the held-out device, which is what a renormalisation factor is.

Model	CV	CV, centred	held out	held out, centred
ridge, log-linear	0.288	0.257	0.442	0.260
random forest	0.209	0.208	0.814	0.427
hist gradient boosting	0.203	0.203	0.950	0.428

Alt text. Error of the three ranked models on the stellarator database under cross-validation and device holdout, each whole and with every device’s mean log residual removed. Centring brings the power law’s holdout error back to its cross-validation level, while the two tree models remain roughly twice their cross-validation error.

ISS04 could not fit these devices with one constant. It carries a renormalisation factor for each device and configuration, and the lock said in advance that a power law refitted on the other devices should therefore be expected to miss a held-out stellarator by a roughly constant factor, whatever its exponents do, which is the reason the outcome could not be called before it was run. That is what happened, and Table S12 shows it. With the device offset removed, the power law’s holdout error is 0.260 against 0.257 under cross-validation: its whole degradation, from 0.288 to 0.442, is the missing constant, and its trends transfer. The forest’s centred error is 0.427 against 0.208, and the booster’s 0.428 against 0.203: granted the same constant, both still fail, so their trends do not transfer. This is also why the power law’s error rises with distance here. The factor ISS04 needs is largest for the devices furthest from the others, and a constant the model cannot know is what the distance is measuring.

S15.9 Secondary arms, interpretation and limitations

None of the secondary arms can change the verdict, and all are reported. At the 10-row threshold, which admits W7-A, the forest is worse on 8 of 8 with $D = 2.49$. With the diamagnetic confinement time as the target for every device, which drops the two devices that have none, it is worse on 5 of 6 with $D = 2.89$. On every measured row, 4847 rows over 8 devices, the forest still wins cross-validation and degrades more, $D = 1.60$, but its holdout mean of 0.688 sits beside the power law’s 0.699 and it is worse on 5 of 8, which under the rule is degradation without a literal inversion. That arm holds the rows the database’s own curators excluded from ISS04, and the power law suffers most from them. The standard set is the primary analysis because the lock said so, not because that arm came out weaker.

What this replication is and is not. It is a qualitative replication of the pattern: the flexible models win the within-device comparison and lose the held-out-device comparison on every device, by the predeclared rule, on a database that shares no device with the one the pattern was found on. It is not a replication of any number, and it is not evidence about tokamaks beyond what HDB5 supplies: these are stellarators and heliotrons, scored on 5 of ISS04’s 6 regressors with no analytic law available on the frozen set, on seven devices whose target is defined by two conventions, with the distance diagnostic behaving differently and for a reason that is itself stellarator-specific. The calibrated-interval collapse, the ITER-size-matched cut and the three-kernel experiment of Sec. S4 were not run here, and nothing in this section speaks to them. The choice of this database was made while a first external replication, on a database that requires an access grant, was still waiting for that grant; its lock stays in force and its outcome will be reported whatever it shows.

S16 Controls on the loss-power identity and on device identity

Section 3.2 of the main text summarises two controls in a table and a paragraph each. This section gives their construction and the numbers behind that table.

The loss power is the target’s denominator. The thermal confinement time is the thermal stored energy over the thermal loss power, so $\log \tau_{\text{th}} = \log W_{\text{th}} - \log P_{\text{LTH}}$ up to the deposit’s dW/dt and radiation conventions, and P_{LTH} is one of the nine features. A log-linear model carries the -1 coefficient on $\log P$ exactly and outside the training range of P , because the relation is definitional; a tree approximates it by axis-aligned splits and, by the bound of Sec. 3.1 of the main text, cannot extrapolate it at all. That admits a reading in which the power law transfers because an identity is exactly log-linear rather than because it has better long-range structure. Refitting every model on the eight remaining features tests it: the direction survives and the strength does not. The forest still wins cross-validation, by a wider margin, 0.198 against 0.315 where it was 0.128 against 0.181, and still loses under leave-one-label-out, but on **10 of 13** labels rather than 13 and **9 of 11** devices rather than 11; the exact tests move to $p = 0.092$ and $p = 0.065$, from values small only under the idealised null of Sec. 2 of the main text, the mean paired gap falls from $+0.251$ to $+0.051$ by label and from $+0.315$ to $+0.070$ by device, C-Mod, JT-60U and JET-ILW change hands, and the gradient booster’s mean gap turns negative. Under both feature sets the model that wins the interpolation comparison is the one that loses the extrapolation comparison, which is the claim Sec. 3 of the main text makes.

The same objection, answered the right way round. Deleting P answers more than the objection asks, since P is a real predictor as well as the target’s denominator. The control the objection calls for regresses $\log W_{\text{th}}$ on the same nine features instead, so that P stays as an ordinary predictor and only the identity goes. STD5 carries no stored energy, but the full DB5.2.3 revision, pinned by SHA-256 and analysed in Sec. S2, carries W_{th} , and joining it to STD5 on machine, shot and time recovers one for **6214** of the 6228 analysed rows; KAPPAA is delivered in both files and agrees across the join to 5×10^{-10} , and on the joined rows $W_{\text{th}}/(\tau_{\text{th}}P)$

has a median of 1.000000. Retargeted, the direction and most of the magnitude survive and the unanimity does not. The forest still wins cross-validation, 0.123 against the power law’s 0.193. Under leave-one-label-out it loses on **9 of 13** labels at a mean gap of +0.220, against 13 of 13 and +0.251 on τ_{th} ; under leave-one-device-out it loses on **9 of 11** at a mean gap of +0.348, larger than the +0.315 the confinement time gives; and across the ITER-size-matched cut it scores **1.081** against the power law’s 0.289, so the collapse the paper rests on is untouched. The exact tests move to $p = 0.267$ and $p = 0.065$. This is the stronger control because the design matrix is unchanged and one coefficient moves: the fitted exponent on $\log P$ goes from -0.688 to +0.308, one unit, the identity carried on the other side rather than taken away, so the slope a tree has to reproduce in that direction more than halves, the tree’s task gets easier, and the gap widens anyway. The specification is in `analysis_stored_energy.py` and `results/stored_energy.json`.

Together the two controls separate what the identity carries from what it does not: it carries the unanimity, since under either control the count falls from thirteen to nine or ten and neither exact test then reaches $p = 0.05$, and it does not carry the effect, since deleting only the identity leaves +0.220 by label and +0.348 by device and the size cut where it was. Two results this paper leans on do not weaken. On a held-out label at nominal 90% the eight-feature conformal intervals of the calibrated-intervals section of the main text cover 42% for the forest and 48% for the booster against 77% for the power law, where the nine-feature figures are 35%, 45% and 83%; and in the kernel comparison of Sec. 6.2 of the main text the bounded squared-exponential rung remains the worst of the three processes, at 0.496 against 0.400 for the linear rung and 0.409 for their sum, where with P it was 0.541 against 0.214 and 0.218, with the linear-kernel process still reproducing ridge to four decimals in both arms. The relation itself is physics rather than a defect in the feature set: the Connor–Taylor constraint on the exponent of P in Sec. 6.1 of the main text is derived from $\tau_E = W/P$, a similarity transformation treating P as an independent scale would produce a constraint no published law satisfies, and the constrained rungs therefore cannot be scored under this ablation at all.

Do the features name the machine? This objection explains everything in Table 1 of the main text without any appeal to long-range behaviour. Four of the nine features barely move once a device is fixed. Major radius carries **0.1%** of its log variance within devices, minor radius 1.1%, inverse aspect ratio 4.9% and elongation 6.1%. A random forest classifier recovers the device from the nine log features at **99.9%** under the same discharge-grouped split used throughout, against a 42.3% majority baseline, so under grouped cross-validation a tree can split on those coordinates and fit a per-device offset: a lookup table, worth a great deal in sample and nothing on a machine that is not in it.

Deleting those features is the obvious control, and it cannot settle the question. Refitting on the four features that do move within a device, plasma current, toroidal field, density and loss power, the reversal disappears: the forest’s interpolation margin *grows* to 50.3% and it is worse than the power law on 6 of 13 labels and 5 of 11 devices, at mean gaps of -0.033 and -0.018, and the cross-validated log-RMSE is 0.157 for the forest against 0.315 for the power law. That is not decisive, because the deleted columns carry the size dependence, which is the thing the power law extrapolates through. The control removes the suspected leak and the physics under test together, and the power law’s held-out score more than doubles, from 0.214 to 0.477: the reversal needs features carrying size, and this cannot say whether because

they carry size or because they name the machine.

The dimensionless coordinates come closer to separating those two things, and they are not coordinates constructed here. Fitting this database in them is established practice on the regression side as well as the experimental one: Cordey *et al.* [18] fit the confinement time against β and collisionality directly, and the reference description of the revision analysed here [9] reports that transforming an engineering-variable scaling into dimensionless form leaves large error bars on the result, proposes regression in the space of dimensionless variables as the alternative, and carries it out. What this arm adds is the use they are put to: a diagnostic on the device-label pathway and not a target-blind control, scored under the held-out-device split against the same models. In $(\rho^*, \beta, \nu^*, q, \epsilon, \kappa, M)$ the size, field and density dependence is retained while no coordinate is a device label: ρ^* moves 26.6% within a device, β 35.4%, ν^* 66.6% and q 70.4%, against major radius's 0.1%. Regressing $B\tau_E$ on those seven groups, with the same models and the same splits, the reversal survives. The forest wins cross-validation by **28.9%**, against 28.7% on the engineering features over the same 6214 rows (the 29% quoted elsewhere is 29.5% on all 6228, rounded), and loses on **11 of 13** labels and **9 of 11** devices, at mean gaps of +0.074 and +0.130; under cross-validation the forest scores 0.229 against the power law's 0.322. The margins are smaller than the engineering-space ones and the direction and the counts are the same.

So features that name the machine account for part of the engineering-space margin and not all of it: a part of that +0.315 is device identity, and +0.130 is what survives in coordinates that carry much less of it. Both numbers are reported, and the dimensionless one is the smaller. Three limits attach to it, and the third decides how it should be read. The temperature it needs comes from the stored energy of the control above, so it is scored on 6214 rows rather than 6228. ϵ and κ still move only 5% and 6% within a device, so the coordinates are cleaner than the engineering ones rather than clean. And that temperature is recovered as $T \sim W_{\text{th}}/(nV)$ rather than measured, while the control above puts $W_{\text{th}}/(\tau_{\text{th}}P)$ at a median of unity on these rows, so the three groups are algebraic functions of the target: substituting the definitions of Eq. (3) of the main text, with $V \sim Ra^2\kappa$, gives $\log(B\tau_E) = \log \beta - \log P + \log V + 3 \log B$ up to that ratio. Two things follow. Neither model is blind in this arm in the way it is in the engineering one, since each is handed a function of the held-out machine's own confinement time. And these coordinates cannot be evaluated for a machine that does not exist, because placing one needs the τ_E being predicted, which is why the forecast of Sec. S3 is stated in engineering variables. This arm is a different experiment carrying a different confound rather than a stricter version of the same one. The specification is in `analysis_dimensionless.py`.

Both objections at once. The two controls above each remove one confound and leave the other in place, and the dimensionless arm that would combine them is not blind to the target. `analysis_joint_control.py` removes both with nothing derived from the target: the target is $\log W_{\text{th}}$ and the features are the four within-device columns, plasma current, toroidal field, density and loss power, every one known before a machine runs. On the 6214 rows that carry a stored energy the forest wins cross-validation, 0.153 against the power law's 0.323, and held out it is level with the power law: worse on 5 of 13 labels and 6 of 11 devices, at mean gaps of +0.002 and +0.050. The same script scores the two single controls on the same rows for comparison, and reproduces their numbers. This arm deletes the size dependence along with the identity, as the within-device arm does, so it cannot say which of the two the reversal

needs; what it shows is that the reversal does not survive the removal of the features that carry both, whichever target is used.

S17 The distance diagnostic and the training-target bound in detail

Section 3.1 of the main text states the distance correlations and the training-target bound in a paragraph each. The uncertainty on those correlations, the per-label detail and the fold-by-fold behaviour of the bound are here.

Thirteen points is few enough that each of those numbers needs an uncertainty. Permuting the distances 20000 times gives $p = 0.0007$ for the forest, $p = 0.060$ for the booster and $p = 0.85$ for the power law. That permutation treats the 13 database labels as exchangeable, which they are not exactly: two physical devices contribute two wall-era labels each, so these p -values are read as Sec. 2 of the main text says: summaries of how lopsided a count is. Recomputed over the 11 physical devices, which removes the duplicated wall-era labels, the forest’s correlation weakens slightly to $+0.77$ ($p = 0.005$) and the power law’s is unchanged at -0.06 ($p = 0.85$), so the contrast the section rests on survives the collapse. That permutation p -value is read the same way. The booster’s falls from $+0.54$ to $+0.38$ ($p = 0.25$). Neither figure clears a conventional threshold, and the booster’s correlation is read throughout as suggestive rather than established. Dropping each label in turn moves the forest’s ρ only within $[+0.81, +0.88]$, so no single extreme device carries it. One control changes how the rest should be read: the *mean baseline*, which fits nothing, also correlates with distance, at $+0.56$. Distance predicts how hard a unit is in general. What is unusual is not that the trees degrade with it but that the power law does not.

Mahalanobis distance between mean log-feature vectors is one measure of how far a unit sits from the training set, and it is the only one used here. It compares first moments through a single pooled covariance, so it is insensitive to differences in the shape or spread of a unit’s operating envelope at a fixed mean, and the covariance it uses is singular by the rank audit of Sec. S12 and so is inverted through the pseudo-inverse. Every statement below about “distance” is a statement about that one diagnostic, not about distributional difference in general.

The per-label scores behind that correlation are in Sec. S6. On the five most distant labels (PBX-M, NSTX, C-Mod, JFT-2M and MAST, all small, two of them spherical) the forest’s error is 2.7 to $4.8\times$ the power law’s; on the four nearest it is within $1.7\times$. JFT-2M and MAST sit within 1% of each other in distance, so the boundary is drawn to be insensitive to that.

A hard bound, and the one fold it binds on. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time anywhere in the training fold, and its largest held-out target is $3.7\times$ above anything any tree in the forest can output. More rows would close this only if they carried larger targets: within the same target range no amount of additional data, and no tuning, moves a ceiling that is a property of the functional form.

That is one fold of thirteen, and where the bound is active is reported here rather than

left to be found, because it decides which reading of Sec. 6.2 of the main text this diagnostic supports. JET is the only held-out label whose targets reach the ceiling at all. On the other twelve the largest held-out target sits below it, by 1.31 in logs on JET-ILW and by 3.27, a factor of 26, on MAST; `results/boundedness.json` carries the margin for every fold. The size cut of Sec. 4 of the main text is the second place it binds, and there **34%** of held-out rows lie above the ceiling.

Set against the per-label gaps, that pattern is evidence, and it points away from the bound rather than toward it. The four labels the forest loses by the widest margins are NSTX (+0.568), PBX-M (+0.453), MAST (+0.427) and JFT-2M (+0.357), and on each of them its ceiling stands about three natural logs above anything the held-out machine reaches. JET, the one fold where the ceiling is in the way, is seventh of the thirteen by margin, at +0.280. The forest therefore fails hardest where the bound is furthest from binding, and a constraint that is not active cannot be what an error is made of. What is left has to operate in the interior of the training range: predictions that stop tracking the features once the input leaves the support the model was fitted on, whether or not a ceiling is anywhere nearby. Sec. 6.2 of the main text builds the model that separates the two readings and finds the second one, and this is the first evidence in the paper that points there.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the argument above does not transfer to it. What can be reported is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 4 of the main text, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

S18 Provenance of every input and number

The Data availability section of the main text names the three third-party inputs. None is redistributed. Each is fetched from its public source by a script in the repository and refused unless its bytes match the pinned fingerprint of Table S13.

Table S13: SHA-256 of each third-party input, verified on load. A file whose bytes differ from these is refused rather than analysed, so every number in this paper refers to one specific set of bytes.

file	SHA-256
<code>hdb5_std5.csv</code>	67601c2da5c51f90cf6298ff499cccc7 4d09ac80c2b98c7dde0d8db3ebb9ac5b
<code>hdb5_db523.csv</code>	7a48e34379663e3e298924990f05cd8f 16b8581516bcfa7bb8f438f83ae80ab6
<code>ISHCDB_26.txt</code>	3a9f857b768ebd601385a9d8e25ac7df 4d48b3403795a43dc484ba6b17fc5f55

Alt text. Table listing the third-party input files, two from the ITPA H-mode confinement database and the stellarator database file, each paired with the SHA-256 fingerprint checked on load. A file whose bytes differ from the printed fingerprint is refused rather than analysed.

What the delivered STD5 file contains, relative to the published count, is settled by matching its rows to the full DB5.2.3 file (`tools/audit_std5_membership.py`, which writes

`results/std5_membership.json`). Every one of the 6228 delivered rows matches a row of the full file on device, shot and time; the deposit's IND column is the full file's row number; every matched row carries the full file's SELDB5 flag; and every one has an ELM My H-mode phase (HGELM, HSELM, HGELMH or HSELMH). The full deposited revision flags 7568 rows, 31 more than the 7537 the database description gives for STD5, so exact identity with the published membership cannot be established from the deposited files. Of the 7568, 1304 have phase H, H-mode with no ELM classification, and none of them is delivered; 6264 are ELM My, of which 6228 are delivered and 36 are not. The 36 are listed in the audit file; no column of the full file separates them from the delivered rows, so the reason they are absent is not recorded in either file. The two devices the published count includes and the delivered file lacks, T-10 and TUMAN-3M, have no ELM My rows. The delivered file is therefore the ELM My H-mode rows of the deposited STD5 selection, the population the ELM My H-mode scalings were fitted to, less 36 rows.

The locked forecast of Sec. S3 also carries a content digest over its input rows, so that a later edit to the operating points is visible:

```
960d537ee3df8780afd2054e51d01a268
f8803aca64a830dba37fd1ff5319192
```

Every script verifies the fingerprint of its input against the pin as it loads the data and refuses to run when the two disagree, so no number in either document can have been computed from a different revision of the database. Several of the JSON outputs under `results/` also record the digest they verified, which carries that provenance into the artefact itself. The analysis code is pinned as firmly as its inputs, at the commit printed in the Data availability section of the main text.

The prose is bound to those artefacts in the same way. The main text states how many of the printed numbers are tied by test to the field of the generated file they came from, and the binding runs in both directions: if an analysis is rerun and its output moves, the rendered value stops matching the printed one and the suite fails; if a number in the text is edited, it stops matching the artefact. A figure that disagrees with the file behind it therefore cannot survive a test run, which is the check a reader would otherwise have to perform by hand. The numbers of the stellarator replication are bound the other way round, because they were written after the rule and not before it: every value the generated file carries is formatted once by a script, every passage about that replication is fenced in the source, and the suite fails on any numeral inside a fence that the script did not produce, so no number in those passages was typed.

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